

ME-Dex 1.0: Bringing Heterogeneous Tactile Sensing into World Action Modeling

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Abstract

World Action Models bring the predictive capabilities of video models into robot action generation, providing a rich foundation for modeling future visual states. Tactile sensing complements this foundation with direct measurements of physical interaction. Some existing methods use tactile features as conditioning inputs without jointly predicting future tactile states, visual observations, and actions. Our key insight is that tactile signals, like video, provide observations of the evolving world state and should be modeled as future observations alongside video. We present ME-Dex-1.0 (MachEmbodied-Dex-1.0), a unified World Action Tactile Model for joint visual, tactile, and action learning. ME-Dex-1.0 adopts a Mixture-of-Transformers architecture comprising a Video Expert, a Tactile Expert, and an Action Expert, all trained with flow matching. We use shared attention connects the experts in intermediate layers, allowing action generation to draw on learned representations of visual and tactile dynamics during joint denoising. To support multi-source heterogeneous tactile inputs, a Canonical Hand Model and a Unified Tactile Autoencoder map tactile observations from different embodiments and sensing layouts into shared spatial and latent spaces. To address the limited availability of paired visual, tactile, and action data, we develop the Agentic Tactile Data Engine, an agent-based data production platform. It supplements RoboTwin and DexJoCo with tactile data recorded directly from force sensors during trajectory replay in simulation. Experiments on the RoboTwin, DexJoCo, and ManiFeel simulation platforms, together with real robot evaluations, demonstrate improved manipulation performance using both grippers and dexterous hands equipped with tactile sensing.

Project Page: <https://machembodied.com/ME-Dex/ME-Dex1.0.html>

Code Repository: <https://github.com/MachEmbodied/ME-Dex-1.0>

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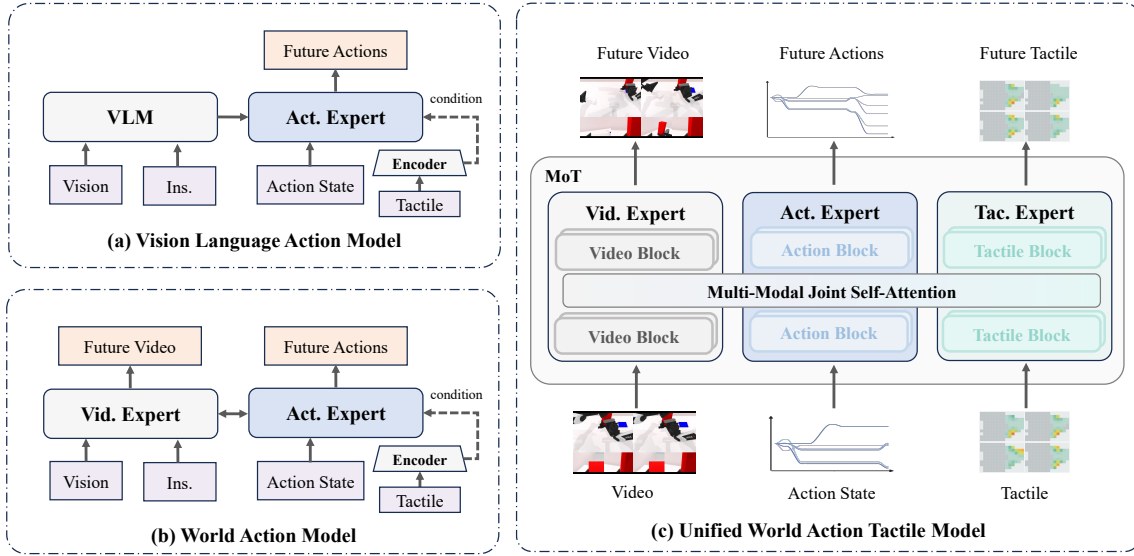


Figure 1. From tactile conditioning to unified world action tactile modeling. (a) Vision-language-action models generate actions from visual observations and task instructions, with some designs additionally conditioning on current tactile observations. (b) World action models jointly predict future video and actions, optionally conditioned on current tactile observations. (c) ME-Dex-1.0 jointly predicts future video, tactile states, and actions within a unified model. The panels illustrate selected modeling designs rather than an exhaustive classification of existing methods.

1 Introduction

Embodied AI aims to build agents that perceive, reason, and act through interaction with the physical world. Recent progress in robot manipulation has followed two closely related directions. Vision-Language-Action models (VLAs) connect visual observations and language instructions to executable robot actions, drawing on knowledge from pretrained vision-language models [1–6]. World Action Models (WAMs) jointly learn robot actions and future visual states, bringing the dynamics learned by video generation models into action prediction [7–12]. As illustrated in Figure 1(a–b), these approaches connect perception to control through direct action generation or joint video-action prediction. Both seek to turn learned representations into effective physical interaction.

Vision provides a rich foundation for this interaction by describing the scene and how objects move. Physical interaction also depends on how the robot applies force to an object and how the object responds. Tactile measurements complement visual observations by describing this interaction. For example, changes in tactile signals during a grasp can help the robot assess whether the object is held securely. Modeling visual and tactile dynamics together therefore offers a path toward a more complete account of the world in which robots act.

Recent works [13–16] have begun to use tactile data in robot policies and world models. A common entry point is to encode tactile data as conditioning inputs, as illustrated in Figure 1(a–b). This offers a straightforward way to add tactile information to existing models and does not require the model to predict how tactile states evolve together with visual states and actions. Recent methods have taken steps toward predictive tactile modeling [17, 18]. Extending this capability across robots requires both scalable tactile data and a shared predictive model that can learn across embodiments.

We aim to build a unified World Action Tactile Model that learns from visual, action, and tactile information across embodiments. Such unification requires a single Model that aligns heterogeneous inputs through a shared Representation and supports joint training on diverse Data at scale. However, realizing this design requires addressing three gaps. The Data Gap arises because datasets such as RoboTwin [19] and DexJoCo [20] lack native tactile measurements, while tactile datasets used in HATO [21] and ManiFeel [22] remain limited in scale and use different data formats. Collecting synchronized tactile and action data also requires effort. The Representation Gap stems from heterogeneous tactile inputs across sensors and embodiments, making them difficult to combine for joint training without a shared representation. The Model Gap concerns how to jointly learn future visual observations, tactile states, and robot actions within a predictive architecture, making future tactile representations accessible to action generation while supporting a shared representation across embodiments.

To address these gaps, we introduce ME-Dex-1.0. To unify data across embodiments and datasets, we design a Canonical Hand Model inspired by the human hand. We map dexterous hands and grippers to this common template and align their tactile measurements. This template establishes a shared spatial reference for heterogeneous tactile inputs. We further propose a Unified Tactile Autoencoder (AE) to encode the aligned tactile data into a shared latent space.

At the model level, we place tactile states on equal footing with video in modeling the future world state. As shown in Figure 1(c), ME-Dex-1.0 adopts a three-expert Mixture-of-Transformers (MoT) architecture. The Video Expert predicts future video as a reference for action generation, while the Tactile Expert predicts future tactile latents that represent the evolution of physical interactions. The Action Expert generates action chunks from the jointly learned representations. Such three experts exchange information through shared attention. We jointly optimize all three experts for future video prediction, future tactile prediction, and action generation. This joint objective trains the Action Expert to draw on representations of future visual observations and tactile states when generating action chunks.

As for paired data, we develop the Agentic Tactile Data Engine, an agent-based platform that records tactile data from force sensors during trajectory replay in simulation. We use this platform to supplement RoboTwin and DexJoCo with tactile data aligned with their visual observations and actions.

Experiments on RoboTwin, DexJoCo, and ManiFeel show that ME-Dex-1.0 outperforms the compared baselines in average manipulation success across grippers and dexterous hands. The results also demonstrate advantages on bimanual tasks and generalization from Clean training to Random evaluation on RoboTwin. To further assess generalization beyond simulation, we evaluate ME-Dex-1.0 on a LeRobot SO-101 with a Paxini PX6AX sensor integrated into its gripper, as well as on Xynova Flex2 dexterous hands. Experiments on both platforms show that the model can execute manipulation tasks using visual and tactile observations.

Overall, our contributions can be summarized as follows:

- We present ME-Dex-1.0, a World Action Tactile Model built on a three-expert MoT architecture. Its Video Expert, Tactile Expert, and Action Expert are jointly optimized to predict future video, future tactile latents, and action chunks, allowing action generation to draw on representations of both future visual and tactile states.
- We propose a Canonical Hand Model inspired by the human hand and a Unified Tactile AE. These components align diverse dexterous hands and grippers to a common template and encode their tactile data into a shared latent space.
- We develop an Agentic Tactile Data Engine for action replay and generation, expanding RoboTwin and DexJoCo with tactile trajectories to support joint visual, tactile, and action learning.

2 Related Work

2.1 Generative Robot Policies and World Action Models

Generative robot policies formulate action prediction as sequence modeling. ACT [23] predicts action chunks with a conditional Transformer, while Diffusion Policy [24] models multimodal action distributions through iterative denoising. Large-scale datasets such as Open X-Embodiment [25] support policy learning across tasks and embodiments. The π_0 [3] family combines a pretrained vision-language backbone with a flow-based action expert, and $\pi_{0.5}$ [26] extends this approach toward open-world generalization. X-VLA [27] explores learning across embodiments through soft prompts. These methods learn action generation from multimodal observations, with future environment states typically represented implicitly.

World Action Models make future visual prediction an explicit part of action learning. Motus [7] combines video generation, action generation, and semantic understanding within a Mixture-of-Transformers architecture. LingBot-VA [10] jointly models video and actions through an autoregressive architecture with causal attention and closed-loop observation updates. The τ_0 -WM model [28] also unifies video and action generation for robot learning. Fast-WAM [29] studies how future video supervision supports action learning when video generation is omitted during inference. MotuBrain [30] further develops unified video-action modeling with modality-specific Transformer streams and shared attention in intermediate layers. These approaches bring the predictive capabilities of video models into robot control. Our work builds on this foundation by modeling future tactile states alongside future visual observations.

2.2 Tactile Conditioned Robot Policies

Tactile observations complement visual perception by providing direct information about physical interaction. Tactile-VLA [16] integrates tactile signals into vision-language-action learning for contact-aware manipulation. FARM [31] conditions a diffusion policy on tactile features and jointly predicts robot pose, gripper width, and gripping force. DECO [32] develops a multimodal diffusion Transformer with separate conditioning pathways for visual, proprioceptive, and tactile inputs. Its plugin tactile adapter introduces tactile features through cross-attention and adapts a pretrained visual policy while keeping the original policy parameters frozen. These designs use observed tactile information to guide action generation. ManiFeel [22] further provides systematic comparisons across tactile modalities, encoders, and policy architectures.

Another line of work uses tactile feedback to refine policy outputs. VLA-Touch [33] combines tactile feedback for high-level planning with a diffusion controller for action refinement. T-Rex [34] adopts a variable-rate Mixture-of-Transformers architecture to combine slower visuomotor processing with faster tactile feedback. Motus2 [35] uses a lightweight tactile expert for action refinement and future force prediction, with an attention structure that prevents action-refinement tokens from attending to noised future-force tokens. These studies explore complementary roles for tactile conditioning, feedback, and predictive supervision. ME-Dex-1.0 focuses on joint visual, tactile, and action modeling, where the Action Expert interacts with intermediate representations of future visual and tactile states during joint denoising.

2.3 Predictive Tactile Modeling

Predictive tactile models learn the expected sensory consequences of physical interaction. Imagine2Touch [36] predicts tactile responses from local visual observations and uses these predictions for object recognition. Visuo-Tactile World Model [37] predicts future visual and tactile states conditioned on robot

actions, supporting physically consistent rollouts and planning. These studies establish tactile observations as prediction targets that describe how robots interact with their environments.

Recent methods connect predictive tactile modeling with action generation and feedback control. OmniVTA [17] combines a visuotactile world model with policy and tactile reflex components. TouchWorld [38] separates subtask planning, visual and tactile goal prediction, action generation, and tactile residual refinement within a hierarchical system. ViTacWorld [39] learns an action-conditioned visual and tactile model for trajectory generation and policy evaluation. VT-WAM [18] jointly learns future visual prediction, tactile deformation prediction, and action prediction through flow matching, using an asymmetric Mixture-of-Transformers architecture and contact-gated attention guidance.

Our work focuses on extending joint predictive modeling to multi-source heterogeneous tactile inputs. ME-Dex-1.0 combines a shared tactile representation across embodiments and sensing layouts with three interacting experts for video, tactile, and action modeling. The Unified Tactile Autoencoder provides a common latent space for current tactile observations and future tactile prediction, while the Mixture-of-Transformers architecture couples these predictions with action generation. This design connects tactile representation across platforms with joint learning of future observations and actions.

2.4 Tactile Representation, Data, and Simulation

Learning from heterogeneous tactile data requires representations that accommodate different sensors and sensing layouts. UniTouch [40] aligns tactile representations with semantic information from other modalities. Sparsh [41] uses self-supervised pretraining to learn general-purpose representations from vision-based tactile data. AnyTouch [42] studies tactile representation learning across sensors and downstream tasks. Sparsh-X [43] extends self-supervised tactile learning to image, audio, motion, and pressure signals from the DIGIT360 sensor. These methods explore shared representations across sensor types and complementary sensing modalities.

ME-Dex-1.0 addresses spatial correspondence across grippers and dexterous hands with different tactile sensing layouts. Our Canonical Hand Model maps tactile observations to corresponding hand regions, providing a shared spatial reference for heterogeneous three-dimensional force fields. A Unified Tactile Autoencoder then maps the aligned observations into a common latent space for tactile conditioning and future prediction. This representation connects local contact information with joint visual, tactile, and action learning across platforms.

Simulation provides a complementary source of tactile training data. TACTO [44] renders vision-based tactile observations for robot learning, while Taxim [45] simulates optical tactile images from contact geometry. DiffTactile [46] provides differentiable contact simulation for tactile manipulation. At the task level, RoboTwin 2.0 [19] supports large-scale bimanual data generation with domain randomization, DexJoCo [20] provides dexterous manipulation environments and demonstrations, and ManiFeel [22] supports systematic studies of visuotactile policies.

To expand paired visual, tactile, and action data, we develop the Agentic Tactile Data Engine, an agent-based data production platform. It augments RoboTwin and DexJoCo, which lack native tactile observations, through trajectory replay in simulation. Tactile data are recorded directly from force sensors and paired with visual observations and actions. Together with the unified tactile representation, these data support joint learning across platforms with different embodiments and sensors.

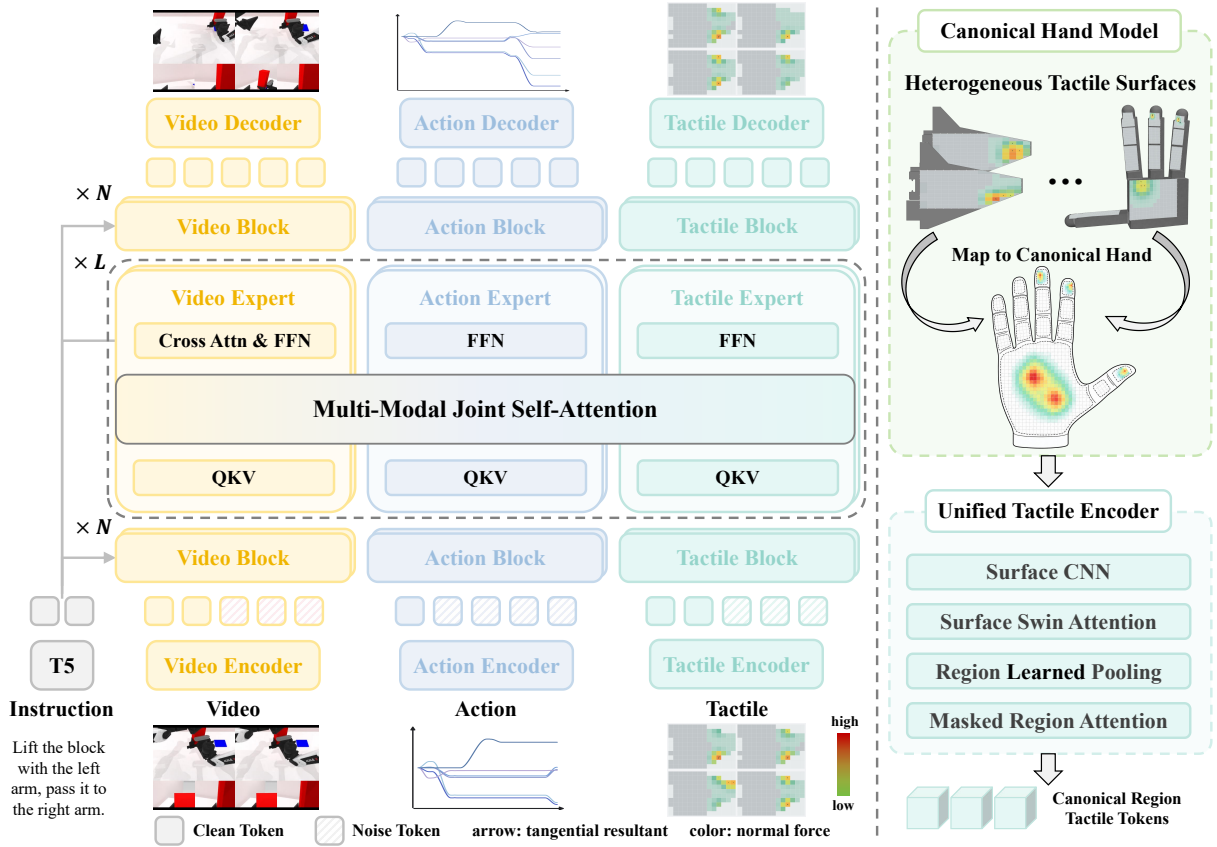


Figure 2. Overview of MachEmbodied-Dex-1.0 The World Action Tactile Model (left) jointly predicts future video, tactile states, and action chunks through three experts trained with conditional flow matching. H-Bridge shared attention connects the Video Expert, Tactile Expert, and Action Expert in intermediate layers, enabling interaction among their evolving representations during joint denoising. Shallow and deep layers retain independent processing. The Unified Tactile Representation (right) aligns heterogeneous tactile measurements through a Canonical Hand Model and uses a shared tactile encoder to produce canonical region tokens.

3 Method

3.1 Overview

As shown in Figure 2, ME-Dex-1.0 jointly predicts future video, tactile states, and action chunks, treating visual and tactile signals as complementary observations of the future world state. This joint prediction allows action generation to draw on representations of both visual and tactile dynamics.

To support multi-source heterogeneous tactile data, the Unified Tactile Representation maps data from different sensors onto a Canonical Hand Model. A shared tactile autoencoder encodes the aligned data into a common latent space while preserving local force structure.

To unify visual and tactile prediction with action generation, the World Action Tactile Model adopts a Mixture-of-Transformers architecture with video, tactile, and action experts jointly trained with flow matching. H-Bridge shared attention connects the experts in intermediate layers, allowing the Action Expert to interact with representations of future visual and tactile states during joint denoising.

To address missing tactile data in existing datasets and support joint visual, tactile, and action training, we develop the Agentic Tactile Data Engine. The engine supplements existing datasets with tactile data directly from force sensors in simulation.

3.2 Problem Formulation

World action models jointly predict future visual observations and actions, allowing representations learned through visual prediction to inform action generation. ME-Dex-1.0 extends this principle to tactile prediction. Our goal is to generate action chunks informed by both future scene evolution and future physical contact.

At time t , the model receives a task instruction ℓ , a visual observation $\mathbf{I}_t$ serving as the first video frame, a proprioceptive state $\mathbf{s}_t$, and the current tactile observation $\mathbf{T}_t$. Visual and tactile encoders map the observations into their respective latent spaces.

$$\mathbf{z}_t^{\text{vis}} = E_{\text{vis}}(\mathbf{I}_t), \quad \mathbf{z}_t^{\text{tac}} = E_{\text{tac}}(\mathbf{T}_t), \quad C_t = (\ell, \mathbf{s}_t, \mathbf{z}_t^{\text{vis}}, \mathbf{z}_t^{\text{tac}}). \quad (1)$$

The tactile observation contains the available signals from both hands, organized according to the canonical representation defined in Section 3.3.

Let $\mathbf{A}^+ = (\mathbf{a}_t, \dots, \mathbf{a}_{t+k-1})$ denote a chunk of k actions, and let $\mathbf{V}^+$ and $\mathbf{T}^+$ denote future video and tactile sequences covering the same physical time window. The superscript $+$ denotes future sequences. We represent the future observable world through two complementary modalities.

$$\mathbf{Z}^{\text{vis},+} = E_{\text{vis}}(\mathbf{V}^+), \quad \mathbf{Z}^{\text{tac},+} = E_{\text{tac}}(\mathbf{T}^+), \quad \mathbf{Z}^{\text{world},+} = (\mathbf{Z}^{\text{vis},+}, \mathbf{Z}^{\text{tac},+}), \quad (2)$$

where E_{tac} operates frame by frame on a tactile sequence. Video describes visible scene evolution, while tactile observations describe the spatial distribution and temporal evolution of physical contact. Both modalities are prediction targets in the joint model.

$$p_{\theta}(\mathbf{A}^+, \mathbf{Z}^{\text{world},+} \mid C_t) = p_{\theta}(\mathbf{A}^+, \mathbf{Z}^{\text{vis},+}, \mathbf{Z}^{\text{tac},+} \mid C_t). \quad (3)$$

The resulting action policy is the action marginal of this joint distribution.

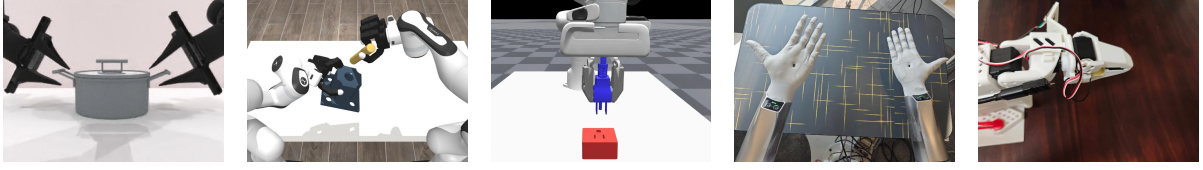
$$\pi_{\theta}(\mathbf{A}^+ \mid C_t) = \int p_{\theta}(\mathbf{A}^+, \mathbf{Z}^{\text{vis},+}, \mathbf{Z}^{\text{tac},+} \mid C_t) d\mathbf{Z}^{\text{vis},+} d\mathbf{Z}^{\text{tac},+}. \quad (4)$$

This marginal is realized by jointly sampling actions and future observation latents and retaining the action output. Future video and tactile states participate in the same generative process as the actions. Their ground-truth values are used as training targets and are unavailable at inference.

To make this dependence part of the computation, the three experts exchange intermediate representations during joint denoising. Let $\mathbf{H}_j^m$ denote the hidden features of modality $m \in \{\text{vis}, \text{tac}, \text{act}\}$ at layer j , and let $\mathcal{B}$ denote the set of H-Bridge layers. The action update at a bridge layer can be expressed abstractly as

$$\mathbf{H}_{j+1}^{\text{act}} = \mathcal{F}_j^{\text{act}}(\mathbf{H}_j^{\text{act}}, \mathbf{H}_j^{\text{vis}}, \mathbf{H}_j^{\text{tac}}), \quad j \in \mathcal{B}, \quad (5)$$

where $\mathcal{F}_j^{\text{act}}$ denotes the action-side Transformer update with shared attention. The visual and tactile features evolve as their experts estimate the corresponding future observations. Action generation therefore interacts with representations of both futures throughout the shared layers and successive denoising steps. This provides a direct computational path through which learning future contact dynamics, alongside visual dynamics, can inform the action prediction.



RoboTwin

DexJoco

ManiFeel

Xynova Flex2

LeRobot SO-101 PX

Figure 3. Multi-source heterogeneous tactile inputs. From left to right, RoboTwin, DexJoCo, ManiFeel, Xynova Flex2 dexterous hands, and a tactile-equipped LeRobot SO-101 gripper. These sources motivate MachEmbodied-Dex-1.0’s unified tactile representation, which expresses heterogeneous contact measurements across embodiments and sensing layouts in a common spatial and latent space.

3.3 Unified Tactile Representation

3.3.1 Tactile Inputs

Our tactile inputs come from diverse grippers and dexterous hands, including grippers in RoboTwin and ManiFeel, dexterous hands in DexJoCo, and Xynova Flex2 dexterous hands, as described in Figure 3. These examples span different sensing surfaces and layouts. To model their contact dynamics with a shared predictor, we need a representation that retains the spatial structure of local contact while providing a common interface across end effectors.

Each sensing surface r provides a force field $\mathbf{F}_t^{(r)} \in \mathbb{R}^{3 \times H \times W}$ containing local normal and two tangential force components on an $H \times W$ grid. A layout mask identifies valid sensing positions, including positions without contact. Preprocessing applies a smooth dead zone, channelwise asinh compression, normalization, and clipping. Normal and tangential components lie in $[0, 1]$ and $[-1, 1]$, respectively.

3.3.2 Canonical Hand Model

To give tactile measurements a consistent spatial meaning across embodiments, we design a Canonical Hand Model inspired by the human hand and treat grippers and dexterous hands as morphological variations of this template. The template defines corresponding finger and palm regions for the left and right hands. Region assignments establish a common spatial reference without requiring identical surface geometry across end effectors.

Figure 2 illustrates the mapping. Each dexterous-hand sensing surface is assigned to its corresponding finger or palm region. The two sensing surfaces of each RoboTwin gripper map to thumb and index regions, with each surface divided into four disjoint, ordered patches. Different sources observe different subsets of the template. A region mask distinguishes unavailable regions from observed regions without contact, preserving the difference between missing measurements and measured absence of contact.

3.3.3 Unified Tactile Autoencoder

Canonical assignment establishes spatial correspondence, but the underlying force fields still differ in resolution, surface geometry, and sensing coverage. A shared future-tactile predictor therefore needs a compact latent representation with consistent spatial organization across these layouts. The representation must also preserve local force patterns and contact transitions, since these are the physical quantities that future prediction should capture. We address these requirements with a Unified Tactile Autoencoder that learns to reconstruct observed force fields from canonical region tokens, providing the tactile counterpart to the latent representation used for video prediction.

The deterministic encoder shares parameters across sensing surfaces and processes each frame inde-

pendently. A residual convolutional network extracts local force features, followed by window and shifted-window attention within each surface. Masked learned pooling aggregates features over valid sensing positions. Gripper surfaces are pooled separately within their four patches, while dexterous-hand surfaces use whole-surface pooling.

The tokens are assigned to canonical slots and augmented with region and pooling-type embeddings. Two masked Transformer layers exchange information among valid regions within a frame, producing $\mathbf{z}_t^{\text{tac}} \in \mathbb{R}^{N \times d}$ with N region tokens of width d . Unobserved regions remain masked and have zero output. Shared parameters and consistent region ordering provide a common latent representation across embodiments, while region tokens retain the spatial organization needed to model local contact evolution.

The decoder gathers the tokens assigned to each surface and uses local grid positions as queries in cross attention. A residual multilayer perceptron and convolutional refinement recover surface features. A force head predicts normal and tangential components, while a contact head predicts contact probabilities. The reconstructed force field is the product of the force prediction, contact probability, and valid-position mask, with the latter two broadcast across force channels.

We train the autoencoder with

$$\mathcal{L}_{\text{AE}} = \mathcal{L}_{\text{force}} + \mathcal{L}_{\text{contact}} + \mathcal{L}_{\text{change}} + \lambda_{\text{lat}} \mathcal{L}_{\text{latent}}. \quad (6)$$

The force loss applies Smooth L1 supervision to force predictions before contact gating at contact positions and to gated reconstructions at background positions. The contact loss uses balanced binary cross entropy with soft labels derived from force magnitudes before dead-zone processing. The change loss matches force differences between adjacent frames, with greater weight on contact onset and release. The latent loss constrains pooled features of valid regions without contact toward those obtained from zero input under the same layout, weighted by λ_{lat} . These terms encourage the representation to preserve force magnitude, contact support, and temporal changes while giving observed non-contact regions a consistent latent reference. After pretraining, the autoencoder is frozen and supplies observed tactile conditions and future tactile targets for the joint model, as well as the decoder for predicted tactile latents.

3.4 World Action Tactile Model

3.4.1 Three-Expert Architecture

The World Action Tactile Model uses a Mixture-of-Transformers architecture with a Video Expert, a Tactile Expert, and an Action Expert. Each expert is a separate Transformer trained with flow matching. The Video Expert follows Wan2.2-5B [47], with 30 Transformer layers and a hidden width of 3072. The Action Expert and Tactile Expert also have 30 layers, with hidden widths of 1024 and 512, respectively. We initialize the Video Expert and Action Expert from Motus weights pretrained on diverse robot data [7] and initialize the Tactile Expert randomly.

During training, the Video Expert receives the encoded first frame together with noisy future video latents; the Action Expert receives the proprioceptive state together with noisy ground-truth actions; and the Tactile Expert receives the encoded current tactile frame together with noisy future tactile latents. The corresponding ground-truth future video, actions, and tactile observations provide supervision through their flow-matching targets. Each expert retains its own representation and hidden width, while their common depth supports interaction at intermediate layers.

The Tactile Expert treats tactile as a temporal observation modality with a structure analogous to video. A future tactile sequence contains T frames, each describing the available tactile signals from both hands at

its corresponding time. Applying the frozen tactile encoder frame by frame produces

$$\mathbf{Z}^{\text{tac},+} = (E_{\text{tac}}(\mathbf{T}_1^+), \dots, E_{\text{tac}}(\mathbf{T}_T^+)) \in \mathbb{R}^{T \times N \times d}. \quad (7)$$

The Tactile Expert is a diffusion Transformer (DiT) that denoises this sequence in latent space, conditioned on the current tactile frame and coupled to the Video and Action Experts. At inference, the future sequence is initialized as Gaussian noise with the same latent shape. After joint denoising, the tactile decoder maps each predicted latent frame back to tactile force fields. This design models tactile observations as a temporal sequence of spatially organized latents. Predicting this sequence brings contact dynamics into world action modeling through the same generative formulation as video.

3.4.2 H-Bridge Shared Attention

Inspired by H-Bridge [48], we restrict cross-modal interaction to an intermediate set of Transformer layers $\mathcal{B}$. Within these layers, the experts project their features into compatible attention dimensions and attend jointly over video, tactile, and action tokens before mapping the outputs back to their own hidden widths. In the shallow and deep layers, each expert processes its tokens independently. This design concentrates information exchange in the middle of the network, preserves modality-specific processing at its input and output stages, and reduces the number of layers requiring cross-modal attention. The Action Expert can thus access representations of both future visual evolution and contact dynamics through the shared layers.

3.4.3 Joint Training and Inference

We train all three experts with conditional flow matching. For modalities $m \in \mathcal{M} = \{\text{vis}, \text{tac}, \text{act}\}$, the clean future targets are

$$\mathbf{x}_0^{\text{vis}} = \mathbf{Z}^{\text{vis},+}, \quad \mathbf{x}_0^{\text{tac}} = \mathbf{Z}^{\text{tac},+}, \quad \mathbf{x}_0^{\text{act}} = \mathbf{A}^+. \quad (8)$$

Each modality independently samples a noise level $\sigma_m \in [0, 1]$ and standard Gaussian noise ϵ^m . We construct noisy targets and their target velocities as

$$\mathbf{x}_{\sigma_m}^m = (1 - \sigma_m)\mathbf{x}_0^m + \sigma_m\epsilon^m, \quad \mathbf{u}^m = \epsilon^m - \mathbf{x}_0^m. \quad (9)$$

The first video frame, proprioceptive state, and current tactile frame remain uncorrupted conditions. Let $\mathbf{X}_\sigma = \{\mathbf{x}_{\sigma_m}^m\}_{m \in \mathcal{M}}$ and $\sigma = \{\sigma_m\}_{m \in \mathcal{M}}$. The experts predict modality-specific velocities while exchanging features through H-Bridge, giving the loss

$$\mathcal{L}_{\text{FM}}^m = \mathbb{E}[\ell_m(\mathbf{v}_\theta^m(\mathbf{X}_\sigma, \sigma, C_t), \mathbf{u}^m)]. \quad (10)$$

The expectation is over demonstration windows, noise levels, and Gaussian noise. For video and actions, ℓ_m is mean squared error. For tactile prediction, it is a weighted squared error that emphasizes changes in ground-truth tactile latents. The joint objective is

$$\mathcal{L}_{\text{joint}} = \lambda_{\text{vis}}\mathcal{L}_{\text{FM}}^{\text{vis}} + \lambda_{\text{tac}}\mathcal{L}_{\text{FM}}^{\text{tac}} + \lambda_{\text{act}}\mathcal{L}_{\text{FM}}^{\text{act}}, \quad (11)$$

where the λ coefficients balance the three prediction tasks. Future visual and tactile supervision shapes the representations exchanged with the Action Expert, while action supervision trains the coupled model to generate the demonstrated controls. Tactile dynamics thus participate in both future observation prediction and the intermediate computation of actions.

At inference, future video latents, future tactile latents, and actions are initialized from Gaussian noise and jointly updated from noise levels of one toward zero using the predicted velocities, under fixed observation

conditions C_t . H-Bridge exchanges information among the evolving modality representations during these updates. The resulting action chunk provides the control output, and the frozen visual and tactile decoders can recover the corresponding predicted observations from their latents.

3.5 Agentic Tactile Data Engine

We develop an Agentic Tactile Data Engine for action replay and generation in simulation. Replaying existing RoboTwin and DexJoCo actions allows force sensors in the simulation to directly record tactile signals for those trajectories. Generating and executing new action trajectories provides additional tactile data. The resulting trajectories pair tactile signals with visual observations and actions over shared time windows, providing both current observation conditions and aligned future targets for joint learning.

4 Experiments

4.1 Experimental Setup

Datasets and Platforms. *RoboTwin.* We augment RoboTwin demonstrations with tactile signals collected directly from force sensors in simulation. We train jointly on 50 tasks using 27,500 Clean and Random trajectories and evaluate under both configurations. An additional setting uses only 2,500 Clean trajectories to assess generalization to Random configurations.

DexJoCo. We augment DexJoCo demonstrations with tactile signals collected directly from force sensors in simulation. Following its official multi-task setting, we jointly train on 1,100 demonstrations spanning six single-arm and five bimanual tasks [20].

ManiFeel. ManiFeel provides three-dimensional tactile force observations [22]. We use 50 demonstrations per task across four insertion tasks, training and evaluating a separate policy for each.

Physical Robots. We qualitatively assess real-world deployment on two robot systems: a single-arm LeRobot SO-101 with a Paxini PX6AX tactile sensor integrated into its gripper, and a dual-arm robot equipped with XR5-10-FS robotic arms and Xynova Flex2 dexterous hands.

Baselines and Evaluation. On RoboTwin, we compare against $\pi_{0.5}$ [26], Fast-WAM [29], and Motus [7]. On DexJoCo, we compare against DP-T [24], $\pi_{0.5}$, DECO and DECO.p [32]. The DP-T results are quoted from DexJoCo [20], the DECO results follow its official benchmark release [32], and the results of DECO.p follows DECO’s official force–tactile approach with its released DexJoCo checkpoint on our tactile-augmented DexJoCo raw data. On ManiFeel, we use the official policy as our baseline and denote it as ManiFeel. To examine the effect of tactile observations on different policies, we construct tactile-conditioned variants of $\pi_{0.5}$ and Fast-WAM, denoted as $\pi_{0.5}+\text{TAC}$ and Fast-WAM+TAC, where we encode tactile input using the pretrained Unified Tactile Autoencoder, for $\pi_{0.5}+\text{TAC}$, we concatenate the resulting tactile tokens with the image and language tokens as prefix tokens that condition the action expert, for Fast-WAM+TAC and ME-Dex-1.0(Tac Cond.) we concat them with the language tokens generated from T5 [49]. The remaining architecture and training settings are unchanged from the corresponding baseline. For the policies trained in this study, we use a single training seed for each experiment. Our primary manipulation metric is closed-loop task success rate, with success determined by the official criteria of each benchmark.

Implementation Details. We initialize the Video Expert and Action Expert of ME-Dex-1.0 from pretrained Motus weights [7] and randomly initialize the Tactile Expert. The Unified Tactile Autoencoder is pretrained separately on tactile data from multiple sources and remains frozen during policy training on each benchmark to prevent the tactile representation from drifting across tasks.

Table 1. Multi-task manipulation performance on RoboTwin. Success rates (%) under Clean + Random training. Train and Eval indicate current tactile observation usage. Zeroed denotes zero tactile input at evaluation. Future tactile prediction denotes training with future tactile targets and predicting future tactile states during inference. VA models video and actions with joint attention at all layers. Tac Cond. adds current tactile conditions, and Full Joint additionally predicts future tactile states. The complete model uses H-Bridge shared attention.

Method	Tactile obs.		Future tactile prediction	H-Bridge	Success rate $\uparrow$	
	Train	Eval			Clean	Random
<i>Baselines</i>						
$\pi_{0.5}$	No	Not used	No	—	52.76	60.74
Fast-WAM	No	Not used	No	—	89.22	89.22
Motus	No	Not used	No	—	88.66	87.02
$\pi_{0.5}$ +TAC	Yes	Observed	No	—	52.60	57.66
Fast-WAM+TAC	Yes	Observed	No	—	90.22	90.22
<i>ME-Dex-1.0 and variants</i>						
ME-Dex-1.0 (VA)	No	Not used	No	No	88.74	86.90
ME-Dex-1.0 (Tac Cond.)	Yes	Observed	No	No	89.04	89.54
ME-Dex-1.0 (Full Joint)	Yes	Observed	Yes	No	89.36	90.60
ME-Dex-1.0	Yes	Observed	Yes	Yes	91.56	91.92
ME-Dex-1.0 (Zero Tac Input)	Yes	Zeroed	Yes	Yes	91.70	91.74

Table 2. Manipulation performance on RoboTwin under Clean training. Baseline results are taken from the RoboTwin leaderboard [19]. Average denotes the mean success rate across the two evaluation configurations. Tactile observations are set to zero for ME-Dex-1.0 under both evaluation configurations.

Method	Success rate (%) $\uparrow$		
	Clean $\rightarrow$ Clean	Clean $\rightarrow$ Random	Average
Fast-WAM	77.8	1.9	39.9
Xiaomi Robotics-0	62.9	18.2	40.5
$\pi_{0.5}$	70.7	46.0	58.4
4D-WAM	81.5	41.8	61.6
GigaBrain-0.7	66.8	67.9	67.3
OLA-Sem	75.1	67.6	71.3
ME-Dex-1.0	89.6	68.1	78.9

4.2 Quantitative Comparison

4.2.1 RoboTwin Benchmark

Mixed Clean and Random Training. Under mixed Clean and Random training, ME-Dex-1.0 achieves success rates of 91.56% and 91.92% in the Clean and Random configurations, respectively, as shown in Table 1. These results exceed Fast-WAM by 2.34 and 2.70 percentage points and Fast-WAM+TAC by 1.34 and 1.70 percentage points.

Within the ME-Dex-1.0 variants, adding current tactile conditions to the video-action model increases Random success from 86.90% to 89.54%. Jointly predicting future tactile states further increases success to 90.60%, and using H-Bridge shared attention raises it to 91.92%. The effect of tactile conditioning varies across external baselines. Under Random evaluation, Fast-WAM gains 1.00 percentage point, whereas $\pi_{0.5}$ loses 3.08 percentage points.

Setting the current tactile input of the trained ME-Dex-1.0 model to zero yields 91.70% on Clean and

Table 3. Multi-task manipulation performance on DexJoCo. Success rates (%) under the *rand-obj* configuration. DP-T, $\pi_{0.5}$, and DECO use no tactile input, whereas $\pi_{0.5}$ +TAC, DECO.p, and ME-Dex-1.0 use tactile data. DECO.p is the tactile-adaptor variant of DECO. /B denotes a bimanual task, and Bimanual avg. is the equal-weight mean across the five /B tasks. DP-T and $\pi_{0.5}$ results are quoted from DexJoCo, and DECO results are taken from its official benchmark release. Bold indicates the highest success rate in each row.

Task	DP-T	$\pi_{0.5}$	$\pi_{0.5}$ +TAC	DECO	DECO.p	ME-Dex-1.0
Hammer Nail	58.7	40.0	10.0	68.0	76.0	86.0
Click Mouse	38.7	76.0	80.0	90.0	80.0	76.0
Pick Bucket	55.3	82.0	46.0	80.0	84.0	92.0
Pinch Tongs	6.0	16.0	0.0	46.0	60.0	92.0
Fold Glasses	11.3	24.0	40.0	50.0	62.0	56.0
Water Plant	60.0	66.0	82.0	88.0	76.0	72.0
Unlock iPad/B	0.0	8.0	0.0	40.0	32.0	48.0
Hanoi/B	8.0	14.0	2.0	18.0	24.0	32.0
Assembly/B	1.3	4.0	6.0	0.0	6.0	28.0
Microwave/B	42.7	70.0	44.0	64.0	68.0	88.0
Photograph/B	28.0	4.0	22.0	76.0	68.0	48.0
Bimanual avg.	16.0	15.7	14.8	39.6	39.6	48.8
Avg.	33.2	36.7	30.2	56.4	57.8	65.3

91.74% on Random, close to the results obtained with observed tactile input. This comparison shows limited sensitivity to current tactile input in this setting. The zero-input variant retains tactile supervision during training and future tactile prediction during inference.

Training on Clean Data Only. We further evaluate generalization from Clean to Random configurations by training ME-Dex-1.0 on only 2,500 Clean demonstrations across the 50 tasks and setting tactile observations to zero during both Clean and Random evaluation. As shown in Table 2, ME-Dex-1.0 achieves success rates of 89.6% on Clean and 68.1% on Random, with an average of 78.9%, exceeding the strongest listed baseline, OLA-Sem, by 6.5 percentage points on average. Its Random success rate is close to the 67.9% reported for GigaBrain-0.7, while its Clean success rate exceeds the strongest listed Clean baseline, 4D-WAM, by 8.1 percentage points.

4.2.2 DexJoCo Benchmark

Under multi-task training across all 11 tasks in the *rand-obj* configuration, ME-Dex-1.0 achieves the highest overall success rate of 65.3%, as shown in Table 3. It outperforms both baselines without tactile input and tactile-conditioned methods, including $\pi_{0.5}$ +TAC and DECO.p, and ranks first on six tasks.

The five bimanual tasks provide tactile data from both hands, offering a useful setting for evaluating joint visual, tactile, and action modeling during coordinated manipulation. ME-Dex-1.0 achieves a bimanual average success rate of 48.8%, compared with 39.6% for both DECO and DECO.p. It obtains the highest success rate on four of the five bimanual tasks, with strong performance on Assembly and Microwave. These results demonstrate its effectiveness across several tasks requiring coordinated contact between both hands, although DECO and DECO.p remain stronger on Photograph.

Among the single-arm tasks, ME-Dex-1.0 achieves the highest success rate on Pick Bucket and Pinch Tongs, reaching 92.0% on both. Its advantages are not uniform across tasks, with $\pi_{0.5}$ leading on Hammer Nail, DECO.p performing better on Fold Glasses and DECO leading on Click Mouse and Water Plant.

Table 4. Manipulation performance on ManiFeel. Success rates (%) with single-task training and 50 evaluation episodes per task.

Method	Peg Insertion	USB Insertion	Power Plug Insertion	Gear Assembly	Average $\uparrow$
ManiFeel	40.0	56.0	58.0	66.0	55.0
ME-Dex-1.0	58.0	74.0	88.0	60.0	70.0

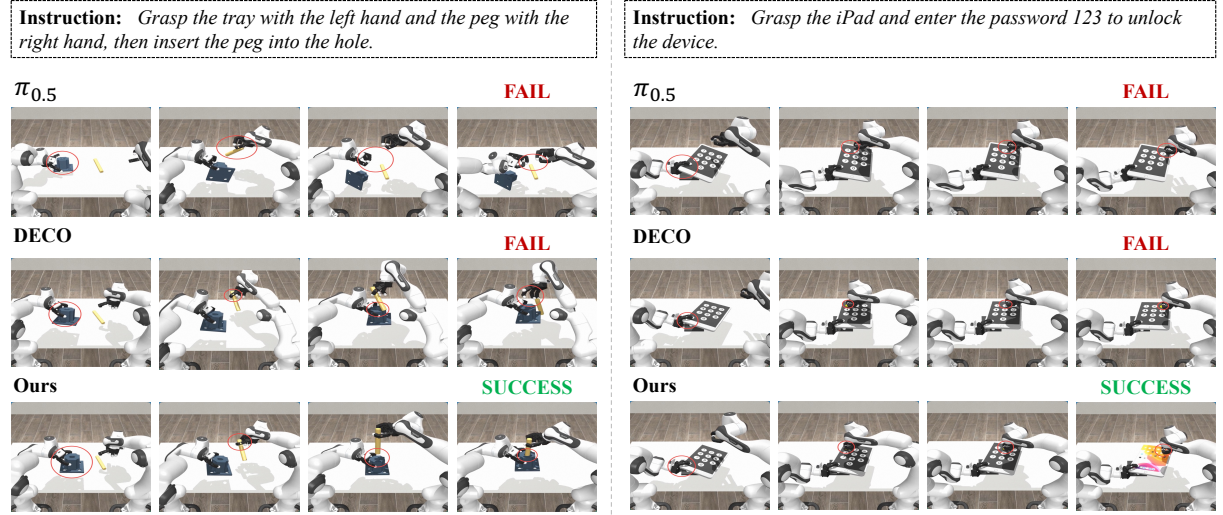


Figure 4. Qualitative comparisons on DexJoCo. Rollouts of $\pi_{0.5}$, DECO, and ME-Dex-1.0 on Assembly (left) and Unlock iPad (right). Frames progress from left to right.

4.2.3 ManiFeel Benchmark

The ManiFeel baseline uses Diffusion Policy to generate action sequences conditioned on visual observations and three-dimensional tactile force fields. ME-Dex-1.0 jointly predicts future video, tactile states, and action chunks, allowing the experts to exchange intermediate representations during prediction.

Table 4 shows that ME-Dex-1.0 outperforms the ManiFeel baseline on three of the four tasks. Its average success rate reaches 70.0%, an improvement of 15.0 percentage points over the baseline. The largest gain occurs on Power Plug Insertion, where success increases from 58.0% to 88.0%. Peg Insertion and USB Insertion each improve by 18.0 percentage points. On Gear Assembly, however, success decreases from 66.0% to 60.0%, indicating that the gains are not uniform across tasks.

4.3 Qualitative Results

4.3.1 DexJoco

Assembly and Unlock iPad examples highlight the benefits of ME-Dex-1.0 on contact-rich bimanual tasks. Both tasks require one hand to stabilize an object while the other performs precise interactions, providing tactile information from both hands. In the figure 4 of shown rollouts, ME-Dex-1.0 exhibits more effective coordination than $\pi_{0.5}$ and DECO during peg insertion and password entry.

4.3.2 Robotwin

ME-Dex-1.0 successfully completes both object scanning and drawer placement in the figure 5 of shown rollouts, coordinating object grasping with subsequent manipulation. In comparison, $\pi_{0.5}$ and Fast-WAM fail in both examples, while Fast-WAM+TAC completes drawer placement but fails the scanning task. These examples highlight the advantage of ME-Dex-1.0 in completing manipulation sequences that require coordinated object interactions.

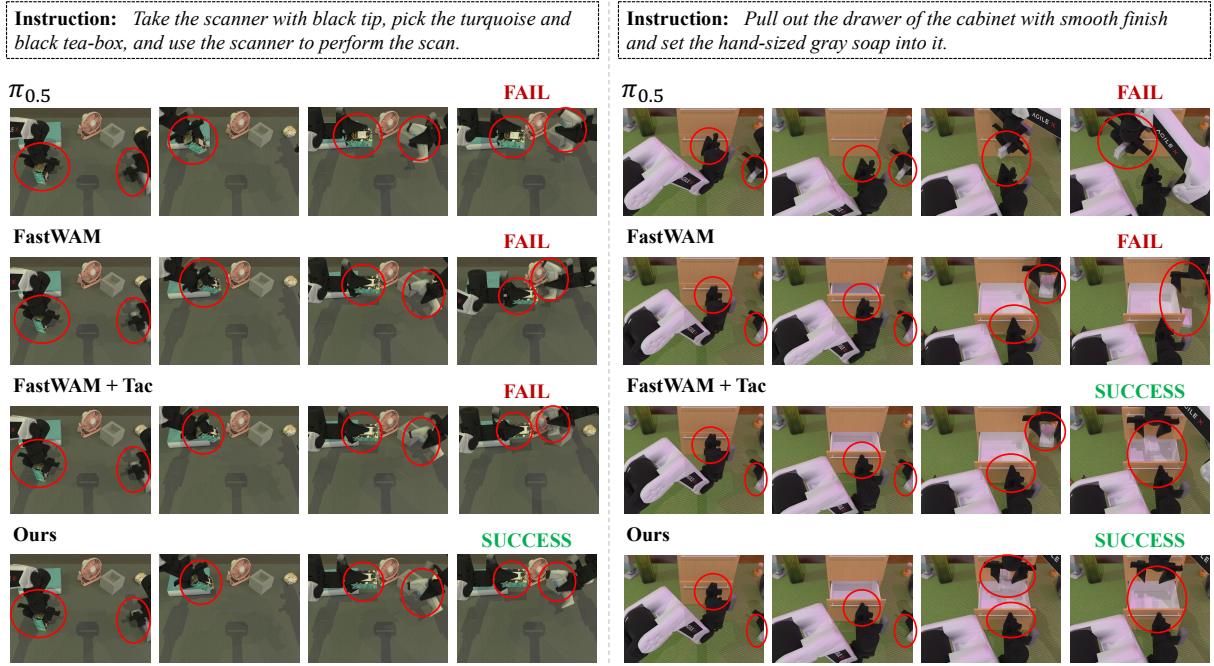


Figure 5. Qualitative comparisons on RoboTwin. Rollouts of $\pi_{0.5}$, Fast-WAM, Fast-WAM+TAC, and ME-Dex-1.0 on object scanning (left) and drawer placement (right). Frames progress from left to right.

Table 5. Tactile reconstruction across platforms. Metrics are averaged equally across subsets within each platform. F1 scores are reported in percent and MAEs in source-specific normalized units.

Platform	Uniformly sampled windows		High-change windows	
	Contact F1 $\uparrow$	Contact force MAE $\downarrow$	Transition F1 $\uparrow$	Transition delta MAE $\downarrow$
RoboTwin	96.56	0.01797	89.73	0.01849
DexJoCo	90.05	0.02807	68.73	0.04293
ManiFeel	99.69	0.05478	66.96	0.06899

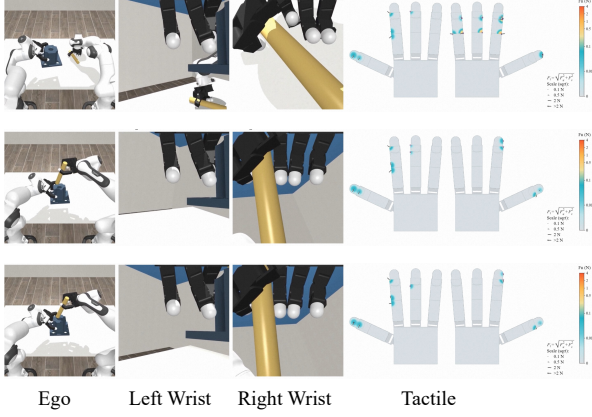
4.4 Unified Tactile Representation

We train a shared Unified Tactile Autoencoder on tactile data from RoboTwin, DexJoCo, and ManiFeel. For each data source, we split trajectories into training and test sets at a ratio of 8 to 2 and combine the training sets for joint training. Evaluation uses uniformly sampled windows and high-change windows. For uniformly sampled windows, we report contact F1 and mean absolute error (MAE) of force values at contact nodes. For high-change windows, we report contact-transition F1 and MAE of frame-to-frame force differences at contact onset and release locations. Subset metrics are averaged equally within each platform, and force errors are computed in the normalized space of the corresponding source.

As shown in Table 5, the shared model achieves contact F1 scores of 96.56%, 90.05%, and 99.69% on RoboTwin, DexJoCo, and ManiFeel, respectively. The contact and force reconstruction results support a shared representation across these sensing layouts. Transition F1 scores of 89.73%, 68.73%, and 66.96% further indicate that the representation retains information about contact onset and release, although the accuracy of these transitions varies across platforms. The force MAEs should be interpreted within each source-specific normalized space.

DexJoco

Instruction: Grasp the tray with the left hand and the peg with the right hand, then insert the peg into the hole.



Robotwin

Instruction: Take the round braided loaf and put it in the pan-shaped skillet with hole in handle.

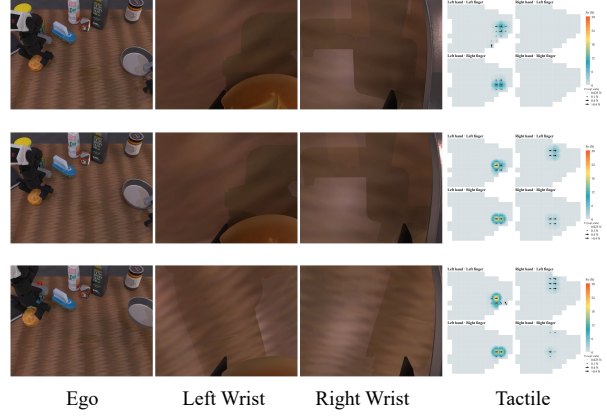
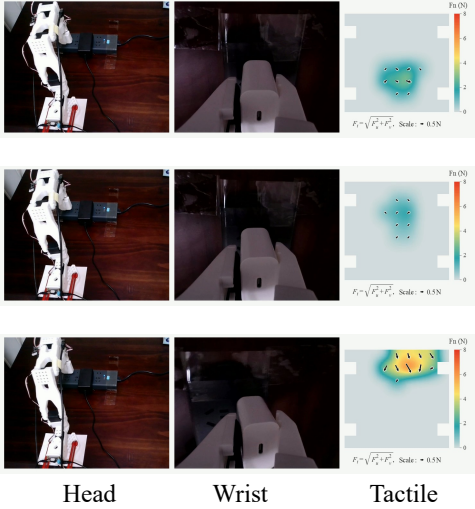


Figure 6. ME-Dex-1.0 rollouts with tactile visualization in simulation. Bimanual assembly on DexJoCo (left) and bread placement on RoboTwin (right). Each row shows ego and wrist-camera views alongside tactile sensor data at a selected timestep. Time progresses from top to bottom.

LeRobot SO-101 PX

Instruction: Move the power plug from one socket to another.



Xynova Flex2

Instruction: Pick up the yellow can on the right with your right hand and stack it upright on top of the other can.

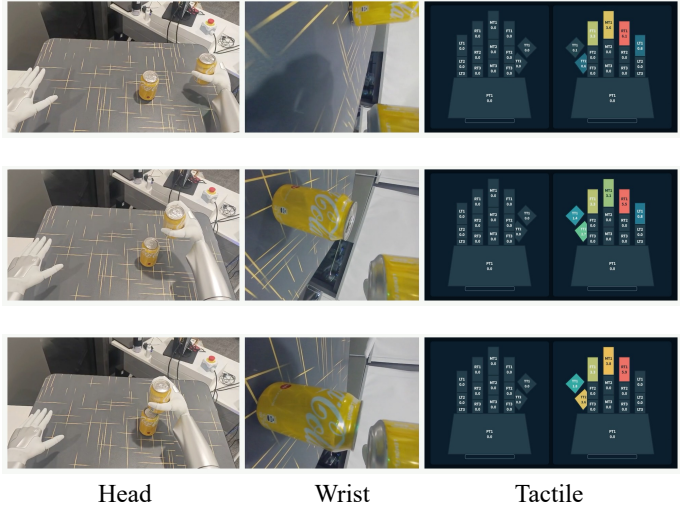


Figure 7. ME-Dex-1.0 rollouts with tactile visualization on physical robots. Power plug transfer with LeRobot SO-101 PX (left) and can stacking with Xynova Flex2 (right). Each row shows head and wrist-camera views alongside tactile sensor data at a selected timestep. Time progresses from top to bottom.

4.5 Rollout Visualization with Tactile Data

We visualize ME-Dex-1.0 rollouts through camera observations and tactile sensor data at selected timesteps. Figure 6 presents examples from DexJoCo and RoboTwin, while Figure 7 shows execution on LeRobot SO-101 PX and Xynova Flex2. The tactile visualizations reveal changes in contact locations and force patterns during object interaction, complementing the corresponding camera views. These examples illustrate inference with multi-source heterogeneous tactile data across grippers and dexterous hands in both simulated and physical environments.

5 Limitation and Future Work

Pretraining of the unified tactile encoder remains limited by data scale and sensor diversity. We will expand cross-platform tactile data to cover more sensing layouts and contact patterns, and systematically evaluate generalization to unseen configurations. Current policies primarily use tactile feedback through observation updates and replanning between action chunks. We will explore tactile-driven local feedback control for high-frequency adjustments to jointly predicted actions during execution, aiming to improve responsiveness to contact changes and precision in fine manipulation.

6 Conclusion

We presented ME-X 1.0, a unified World Action Tactile Model that treats tactile states alongside video as future observations for action generation. Its Mixture-of-Transformers architecture brings together a Video Expert, a Tactile Expert, and an Action Expert for joint learning through flow matching. The experts exchange representations during joint prediction, connecting anticipated visual changes and contact dynamics with action generation. A unified tactile representation aligns multi-source heterogeneous tactile observations across embodiments and sensing layouts, while the Agentic Tactile Data Engine supplies paired training data through simulation. Experiments on RoboTwin, DexJoCo, and ManiFeel demonstrate improved manipulation performance with grippers and dexterous hands.

7 Contributions

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